

1) Для заданной на схеме schema-lab4 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров выполнить планирование и документирование адресного пространства в подсетях LAN1, LAN2, LAN3 и назначить статические адреса маршрутизаторам и динамическое конфигурирование адресов для VPC

У LAN1 будет сеть 192.168.1.0/24 и адрес маршрутизатора 192.168.1.1.

У LAN2 будет сеть 192.168.2.0/24 и адрес маршрутизатора 192.168.2.1.

У LAN3 будет сеть 10.0.0.0/24 и адрес маршрутизатора 10.0.0.1.

Маршрутизатор 1:

```
R1#enable
```

```
R1#conf t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R1(config)#int fa0/0
```

```
R1(config-if)#ip address 192.168.1.1 255.255.255.0
```

```
R1(config-if)#ip helper-address 10.0.0.2
```

```
R1(config-if)#no sh
```

```
R1(config-if)#
```

```
*Mar 1 00:11:04.999: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
```

```
*Mar 1 00:11:05.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface  
FastEthernet0/0, changed state to up
```

```
R1(config-if)#exit
```

```
R1(config)#int fa2/0
```

```
R1(config-if)#ip address 192.168.2.1 255.255.255.0
```

```
R1(config-if)#ip helper-address 10.0.0.2
```

```
R1(config-if)#no sh
```

```
R1(config-if)#
```

```
*Mar 1 00:12:47.979: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
```

*Mar 1 00:12:48.979: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up

R1(config-if)#exit

R1(config)#int fa1/0

R1(config-if)#ip address 10.0.0.1 255.255.255.0

R1(config-if)#no sh

R1(config-if)#

*Mar 1 00:13:44.167: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:13:45.167: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R1(config-if)#exit

R1(config)#exit

Маршрутизатор 2:

R2#enable

R2#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#int fa0/0

R2(config-if)#ip address 10.0.0.2 255.255.255.0

R2(config-if)#no sh

R2(config-if)#

*Mar 1 00:01:48.515: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:01:49.515: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R2(config-if)#exit

R2(config)#exit

2) Настроить сервер DHCP на маршрутизаторе R2 для обслуживания адресных пулов адресного пространства подсетей LAN1 и LAN2

```
R2#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
R2(config)#service dhcp
```

```
R2(config)#ip dhcp excluded-address 192.168.1.1
```

```
R2(config)#ip dhcp excluded-address 192.168.2.1
```

```
R2(config)#ip dhcp pool LAN1
```

```
R2(dhcp-config)#network 192.168.1.0 255.255.255.0
```

```
R2(dhcp-config)#default-router 192.168.1.1
```

```
R2(dhcp-config)#exit
```

```
R2(config)#ip dhcp pool LAN2
```

```
R2(dhcp-config)#network 192.168.2.0 255.255.255.0
```

```
R2(dhcp-config)#default-router 192.168.2.1
```

```
R2(dhcp-config)#exit
```

```
R2(config)#exit
```

3) Настроить статическую (nb!) маршрутизацию между подсетями

На маршрутизаторе 2:

```
R2#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
R2(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

```
R2(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.1
```

```
R2(config)#exit
```

На маршрутизаторе 1 все сети подключены напрямую

4) Проверить работоспособность протокола DHCP и маршрутизации, выполнив ping между всеми VPC

Получение IP:

PC1> ip dhcp

DDORA IP 192.168.1.2/24 GW 192.168.1.1

PC2> ip dhcp

DDORRA IP 192.168.1.3/24 GW 192.168.1.1

PC3> ip dhcp

DDORA IP 192.168.2.2/24 GW 192.168.2.1

PC4> ip dhcp

DDORA IP 192.168.2.3/24 GW 192.168.2.1

Пинги:

PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=14.445 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=2.269 ms

84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=4.907 ms

84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=0.732 ms

84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=0.862 ms

PC1> ping 192.168.2.2

84 bytes from 192.168.2.2 icmp_seq=1 ttl=63 time=29.665 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=63 time=25.041 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=63 time=15.470 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=63 time=15.451 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=63 time=15.119 ms

PC1> ping 192.168.2.3

84 bytes from 192.168.2.3 icmp_seq=1 ttl=63 time=30.196 ms
84 bytes from 192.168.2.3 icmp_seq=2 ttl=63 time=15.342 ms
84 bytes from 192.168.2.3 icmp_seq=3 ttl=63 time=15.204 ms
84 bytes from 192.168.2.3 icmp_seq=4 ttl=63 time=17.876 ms
84 bytes from 192.168.2.3 icmp_seq=5 ttl=63 time=12.938 ms

PC2> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=8.635 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=6.414 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=0.739 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=5.815 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=0.733 ms

PC2> ping 192.168.2.2

84 bytes from 192.168.2.2 icmp_seq=1 ttl=63 time=19.175 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=63 time=14.533 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=63 time=15.702 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=63 time=15.969 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=63 time=14.633 ms

PC2> ping 192.168.2.3

84 bytes from 192.168.2.3 icmp_seq=1 ttl=63 time=11.450 ms

84 bytes from 192.168.2.3 icmp_seq=2 ttl=63 time=15.643 ms

84 bytes from 192.168.2.3 icmp_seq=3 ttl=63 time=15.300 ms

84 bytes from 192.168.2.3 icmp_seq=4 ttl=63 time=15.386 ms

84 bytes from 192.168.2.3 icmp_seq=5 ttl=63 time=15.311 ms

PC3> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=63 time=25.239 ms

84 bytes from 192.168.1.2 icmp_seq=2 ttl=63 time=15.778 ms

84 bytes from 192.168.1.2 icmp_seq=3 ttl=63 time=14.670 ms

84 bytes from 192.168.1.2 icmp_seq=4 ttl=63 time=17.968 ms

84 bytes from 192.168.1.2 icmp_seq=5 ttl=63 time=12.757 ms

PC3> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=63 time=30.230 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=63 time=15.311 ms

84 bytes from 192.168.1.3 icmp_seq=3 ttl=63 time=15.376 ms

84 bytes from 192.168.1.3 icmp_seq=4 ttl=63 time=15.731 ms

84 bytes from 192.168.1.3 icmp_seq=5 ttl=63 time=15.816 ms

PC3> ping 192.168.2.3

84 bytes from 192.168.2.3 icmp_seq=1 ttl=64 time=0.639 ms

84 bytes from 192.168.2.3 icmp_seq=2 ttl=64 time=8.301 ms

84 bytes from 192.168.2.3 icmp_seq=3 ttl=64 time=5.549 ms
84 bytes from 192.168.2.3 icmp_seq=4 ttl=64 time=9.001 ms
84 bytes from 192.168.2.3 icmp_seq=5 ttl=64 time=0.665 ms

PC4> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=63 time=28.597 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=63 time=14.727 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=63 time=19.380 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=63 time=11.572 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=63 time=15.914 ms

PC4> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=63 time=22.070 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=63 time=15.302 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=63 time=15.337 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=63 time=16.050 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=63 time=14.116 ms

PC4> ping 192.168.2.2

84 bytes from 192.168.2.2 icmp_seq=1 ttl=64 time=0.573 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=64 time=0.619 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=64 time=0.711 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=64 time=1.057 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=64 time=3.570 ms

5) Перехватить в Wireshark диалог одного из VPC с сервером DHCP, разобрать с комментариями

От PC1 к коммутатору:

dhcp						
No.	Time	Source	Destination	Protocol	Length	Info
29	40.045719	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x5c0c3566
30	41.045810	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x5c0c3566
33	42.778200	192.168.1.1	192.168.1.2	DHCP	342	DHCP Offer - Transaction ID 0x5c0c3566
35	44.046191	0.0.0.0	255.255.255.255	DHCP	406	DHCP Request - Transaction ID 0x5c0c3566
36	44.065781	192.168.1.1	192.168.1.2	DHCP	342	DHCP ACK - Transaction ID 0x5c0c3566

У Discover и Request адрес отправителя нулевой, а адрес получателя широковещательный, потому что собственный адрес и адрес DHCP-сервера неизвестны ещё. Пакеты доходят по MAC-адресам:

>	Frame 30: Packet, 406 bytes on wire (3248 bits), 406 bytes captured (3248 bits) on interface -, id 0
▼	Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
>	Destination: Broadcast (ff:ff:ff:ff:ff:ff)
>	Source: Private_66:68:00 (00:50:79:66:68:00)
	Type: IPv4 (0x0800)
	[Stream index: 3]
>	Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
>	User Datagram Protocol, Src Port: 68, Dst Port: 67
▼	Dynamic Host Configuration Protocol (Discover)
	Message type: Boot Request (1)
	Hardware type: Ethernet (0x01)
	Hardware address length: 6
	Hops: 0
	Transaction ID: 0x5c0c3566
	Seconds elapsed: 0
>	Bootp flags: 0x0000 (Unicast)
	Client IP address: 0.0.0.0
	Your (client) IP address: 0.0.0.0
	Next server IP address: 0.0.0.0
	Relay agent IP address: 0.0.0.0
	Client MAC address: Private_66:68:00 (00:50:79:66:68:00)
	Client hardware address padding: 00000000000000000000
	Server host name not given
	Boot file name not given
	Magic cookie: DHCP
>	Option: (53) DHCP Message Type (Discover)
>	Option: (12) Host Name

У Discover MAC-адрес получателя широковещательный.

```

> Frame 35: Packet, 406 bytes on wire (3248 bits), 406 bytes captured (3248 bits) on interface -, id 0
▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: cc:01:47:86:00:00 (cc:01:47:86:00:00)
  > Destination: cc:01:47:86:00:00 (cc:01:47:86:00:00)
  > Source: Private_66:68:00 (00:50:79:66:68:00)
  Type: IPv4 (0x0800)
  [Stream index: 5]
> Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
> User Datagram Protocol, Src Port: 68, Dst Port: 67
▼ Dynamic Host Configuration Protocol (Request)
  Message type: Boot Request (1)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x5c0c3566
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
  Client IP address: 192.168.1.2
  Your (client) IP address: 0.0.0.0
  Next server IP address: 0.0.0.0
  Relay agent IP address: 0.0.0.0
  Client MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  > Option: (53) DHCP Message Type (Request)

```

У Request MAC-адрес получателя (сервера) уже известен.

```

> Frame 33: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface -, id 0
▼ Ethernet II, Src: cc:01:47:86:00:00 (cc:01:47:86:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)
  > Destination: Private_66:68:00 (00:50:79:66:68:00)
  > Source: cc:01:47:86:00:00 (cc:01:47:86:00:00)
  Type: IPv4 (0x0800)
  [Stream index: 5]
> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.2
> User Datagram Protocol, Src Port: 67, Dst Port: 68
▼ Dynamic Host Configuration Protocol (Offer)
  Message type: Boot Reply (2)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x5c0c3566
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 192.168.1.2
  Next server IP address: 0.0.0.0
  Relay agent IP address: 192.168.1.1
  Client MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  > Option: (53) DHCP Message Type (Offer)
  > Option: (54) DHCP Server Identifier (10.0.0.2)

```

```

> Frame 36: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface -, id 0
▼ Ethernet II, Src: cc:01:47:86:00:00 (cc:01:47:86:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)
  > Destination: Private_66:68:00 (00:50:79:66:68:00)
  > Source: cc:01:47:86:00:00 (cc:01:47:86:00:00)
  Type: IPv4 (0x0800)
  [Stream index: 5]
> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.2
> User Datagram Protocol, Src Port: 67, Dst Port: 68
▼ Dynamic Host Configuration Protocol (ACK)
  Message type: Boot Reply (2)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x5c0c3566
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
  Client IP address: 192.168.1.2
  Your (client) IP address: 192.168.1.2
  Next server IP address: 0.0.0.0
  Relay agent IP address: 192.168.1.1
  Client MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  > Option: (53) DHCP Message Type (ACK)
  > Option: (54) DHCP Server Identifier (10.0.0.2)

```

У Offer и ACK адрес маршрутизатора в LAN1 является адресом отправителя, а адресом получателя является предлагаемый сервером адрес.

От коммутатора к маршрутизатору 1:

dhcp						
No.	Time	Source	Destination	Protocol	Length	Info
28	30.966226	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x5c0c3566
29	31.965786	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x5c0c3566
32	33.697414	192.168.1.1	192.168.1.2	DHCP	342	DHCP Offer - Transaction ID 0x5c0c3566
34	34.966229	0.0.0.0	255.255.255.255	DHCP	406	DHCP Request - Transaction ID 0x5c0c3566
35	34.985100	192.168.1.1	192.168.1.2	DHCP	342	DHCP ACK - Transaction ID 0x5c0c3566

Пакеты идентичны, так как коммутатор не занимается маршрутизацией.

dhcp						
No.	Time	Source	Destination	Protocol	Length	Info
8	21.324744	192.168.1.1	10.0.0.2	DHCP	406	DHCP Discover - Transaction ID 0x5c0c3566
11	23.053165	10.0.0.2	192.168.1.1	DHCP	342	DHCP Offer - Transaction ID 0x5c0c3566
12	24.332705	192.168.1.1	10.0.0.2	DHCP	406	DHCP Request - Transaction ID 0x5c0c3566
13	24.340389	10.0.0.2	192.168.1.1	DHCP	342	DHCP ACK - Transaction ID 0x5c0c3566

Маршрутизатор 1 подменил адрес отправителя на свой и адрес получателя на адрес маршрутизатора 2 и наоборот в ответных пакетах.

```

> Frame 8: Packet, 406 bytes on wire (3248 bits), 406 bytes captured (3248 bits) on interface -, id 0
▼ Ethernet II, Src: cc:01:47:86:00:10 (cc:01:47:86:00:10), Dst: cc:02:47:a4:00:00 (cc:02:47:a4:00:00)
  > Destination: cc:02:47:a4:00:00 (cc:02:47:a4:00:00)
  > Source: cc:01:47:86:00:10 (cc:01:47:86:00:10)
  Type: IPv4 (0x0800)
  [Stream index: 3]
> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 10.0.0.2
> User Datagram Protocol, Src Port: 67, Dst Port: 67
▼ Dynamic Host Configuration Protocol (Discover)
  Message type: Boot Request (1)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 1
  Transaction ID: 0x5c0c3566
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 0.0.0.0
  Next server IP address: 0.0.0.0
  Relay agent IP address: 192.168.1.1
  Client MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  > Option: (53) DHCP Message Type (Discover)

> Frame 11: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface -, id 0
▼ Ethernet II, Src: cc:02:47:a4:00:00 (cc:02:47:a4:00:00), Dst: cc:01:47:86:00:10 (cc:01:47:86:00:10)
  > Destination: cc:01:47:86:00:10 (cc:01:47:86:00:10)
  > Source: cc:02:47:a4:00:00 (cc:02:47:a4:00:00)
  Type: IPv4 (0x0800)
  [Stream index: 3]
> Internet Protocol Version 4, Src: 10.0.0.2, Dst: 192.168.1.1
> User Datagram Protocol, Src Port: 67, Dst Port: 67
▼ Dynamic Host Configuration Protocol (Offer)
  Message type: Boot Reply (2)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x5c0c3566
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 192.168.1.2
  Next server IP address: 0.0.0.0
  Relay agent IP address: 192.168.1.1
  Client MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  > Option: (53) DHCP Message Type (Offer)
  > Option: (54) DHCP Server Identifier (10.0.0.2)

```

6) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств